FORAM SONI

Agentic AI Developer • Android Developer (5 yrs)

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PROFESSIONAL SUMMARY

Motivated Agentic AI Developer with hands-on project experience designing and implementing autonomous multi-agent systems, Retrieval-Augmented Generation (RAG) pipelines, and LLM-powered applications. Proficient in LangGraph, LangChain, and CrewAI with practical knowledge of ReAct, Corrective RAG, Adaptive RAG, and Self-RAG agentic patterns. Holds a Professional Certificate in Generative AI & Machine Learning from E&ICT Academy, IIT Kanpur and has built end-to-end agentic AI projects. Also brings 5+ years of professional Android development experience (Kotlin, Jetpack Compose, MVVM) delivering production apps across IoT, health, e-commerce, fintech, and supply-chain domains.

TECHNICAL SKILLS

Agentic AI & Frameworks: LangGraph, LangChain, CrewAI, OpenAI Agents SDK, ReAct Pattern, Multi-Agent Orchestration, Agent Memory & State Management, Tool-Using LLMs, Human-in-the-Loop Design, OpenAI Function Calling, Agent Observability (LangSmith)

RAG & Vector Search: Retrieval-Augmented Generation (RAG), Adaptive RAG, Corrective RAG, Self-RAG, Agentic RAG, FAISS, Pinecone, ChromaDB, Semantic Chunking, Hybrid Search, Re-ranking, Embeddings

LLM & AI/ML: Large Language Models (LLMs), Prompt Engineering, Chain-of-Thought, HuggingFace Transformers, PyTorch, TensorFlow, Scikit-learn, LSTMs, GRUs, CNNs, Text Summarisation, TFLite / ONNX

GenAI Tooling & Deployment: OpenAI API (GPT-4o / GPT-3.5), Groq API, Streamlit, FastAPI, Flask, Python, Pandas, NumPy, Jupyter Notebook, Google Colab, MLflow, Weights & Biases

Android & Mobile: Kotlin, Java, Jetpack Compose, Android SDK, MVVM / Clean Architecture, Dagger Hilt, Coroutines, Room, Retrofit, Volley, ARCore, BLE, FCM, Stripe, RazorPay, QPay

Developer Tools & Cloud: Git, GitHub, Bitbucket, GitLab, VS Code, Postman, LangSmith, Trello, Asana

AGENTIC AI PROJECTS

Autonomous Research & Report Generation Agent

LangGraph · LangChain · OpenAI · Tavily Search · Streamlit

- Built a stateful multi-step research agent using LangGraph state machines; agent autonomously plans research, dispatches Tavily web-search tool calls, synthesises findings, and self-corrects on retrieval failure without human prompting.
- Implemented event-driven graph transitions (planning → retrieval → reflection → report-writing nodes) enabling multi-hop reasoning over complex, open-ended queries.
- Applied LangGraph checkpointing and LangChain message history for persistent cross-turn memory, eliminating redundant re-fetching across research iterations.
- Deployed as a Streamlit app exposing per-node reasoning traces for full agent transparency; integrated LangSmith for observability and debugging.

Multi-Agent Blog Content Writing Crew

CrewAI · LangChain · OpenAI · DuckDuckGo Search · Python

- Designed a three-agent CrewAI crew (Planner → Writer → Editor) that autonomously researches a topic, drafts a structured blog post, and self-edits for accuracy and clarity without manual intervention.
- Configured agent roles, goals, and backstories in CrewAI YAML with sequential task chaining and structured output passing between agents via Pydantic models.
- Integrated DuckDuckGo Search and a custom YouTube caption extraction tool, enabling the Planner to retrieve live web data and transcript-grounded content for downstream agents.

Corrective RAG & Self-RAG Pipeline with LangGraph

LangGraph · LangChain · OpenAI · FAISS · Tavily · Python

- Implemented Corrective RAG using LangGraph: agent retrieves context, grades each document for relevance with an LLM evaluator, and triggers query rephrasing + web search when documents are insufficient — eliminating hallucination on under-specified inputs.
- Built Self-RAG variant where the agent autonomously evaluates its own generated answer, decides whether additional retrieval is needed, and re-queries with a reformulated prompt if confidence is below threshold.
- Designed conditional LangGraph edges to route between retrieval, grading, rephrasing, web-search, and generation nodes; the system adapts its full retrieval strategy dynamically per query.

Adaptive RAG System with Query Routing

LangGraph · LangChain · OpenAI · ChromaDB · Tavily · Python

- Built an Adaptive RAG system that classifies incoming queries (simple factual / complex multi-hop / out-of-scope) and routes them to the appropriate pipeline — direct retrieval, multi-step RAG, or live web search.
- Implemented a query analysis router node in LangGraph that prevents over-engineering simple requests while ensuring complex queries trigger

full multi-step retrieval with intermediate reasoning.

- Integrated ChromaDB as a persistent vector store with semantic chunking and hybrid BM25 + dense search for improved retrieval precision across heterogeneous document corpora.

Multi-Agent System Built from Scratch with LangGraph

LangGraph · LangChain · OpenAI · Streamlit · Python

- Architected a multi-agent collaborative system using raw LangGraph primitives — defined specialised agent nodes (researcher, analyst, summariser), directed graph edges, and shared typed state schema without high-level wrappers.
- Implemented message-passing patterns and shared goals between agents, enabling coordinated multi-hop task completion across independent reasoning nodes.
- Added human-in-the-loop interrupt nodes at critical decision points, allowing users to review and approve agent actions before execution continues.

End-to-End SQL Agent with Natural Language Interface

LangChain · LangGraph · OpenAI · SQLite / PostgreSQL · Streamlit

- Built a tool-using SQL agent that converts natural language queries into validated SQL, executes them against a live database, interprets results, and generates human-readable responses — enabling non-technical users to query data conversationally.
- Implemented schema-aware prompt construction and SQL validation guardrails to prevent destructive queries; agent retries with a corrected query if execution fails.
- Deployed as a Streamlit interface with query history, SQL preview, and result table display, making the agent accessible without any SQL knowledge.

AI-Powered Customer Support Multi-Agent System

LangGraph · LangChain · OpenAI · FAISS · FastAPI · Python

- Designed a multi-agent customer support system with specialised agents for query classification, sentiment analysis, knowledge-base retrieval, and smart escalation to human support when confidence is low.
- Built a LangGraph-orchestrated workflow with conditional routing: routine queries answered via RAG retrieval from a FAISS knowledge base; complex or negative-sentiment queries routed to an escalation agent that generates a structured support ticket.
- Exposed the system via a FastAPI backend with session-based conversation memory, enabling stateful multi-turn support interactions without context loss between turns.

PROFESSIONAL EXPERIENCE — ANDROID DEVELOPMENT

Mindinventory

May 2022 – Oct 2023

Software Engineer

Technologies: Kotlin, Jetpack Compose, Coroutines, Retrofit, MVVM with Clean Architecture, Dagger Hilt, Wifi Scanning, ARCore

- Collaborated with team members to gather requirements, provide technical expertise, and ensure project alignment with business objectives.
- Led design of Native UI for a Health & Lifestyle Application using Jetpack Compose, implementing hotel Wi-Fi connectivity, deals/offers retrieval, Exoplayer-based exercise video streaming, and activity progress charts.
- Contributed to an in-house E-Commerce app incorporating AR-Core for Augmented Reality product display and custom Jetpack Compose UI.
- Built in-house MVVM Clean Architecture project templates and a shared Jetpack Compose component repository to accelerate fellow developers.
- App link: play.google.com/store/apps/details?id=com.app.gympak

Excellent Web World

Oct 2020 – May 2022

Android Developer

Technologies: Kotlin, Retrofit, FCM, RazorPay, Room DB, MVVM, Dagger Hilt, Coroutines, QPay Payment Gateway

- Developed a full Food Delivery application (customer + delivery boy) with offline cart management, QPay & Wallet payment gateways, Google Maps integration, order status tracking, and real-time Chat via WebSocket.
- Contributed to a Parental Control application to block specified apps on children's devices until assigned tasks are completed.
- Implemented BLE device connectivity for a Smart Parking App and conducted research in Indoor Navigation.
- Refactored legacy Java codebase to Kotlin, improving readability and maintainability, resulting in a measurable reduction in development time.
- Collaborated cross-functionally to gather requirements, define project scope, and ensure timely delivery.
- Contributed to sound-reactive hardware integration with mobile devices via Bluetooth.
- App link: play.google.com/store/apps/details?id=com.app.hyperspace

Avancer Technologies Ltd

Jun 2018 – Oct 2019

Software Engineer

Technologies: Android, Firebase Realtime Database, WebSockets, Barcode Scanner, Charts

- Maintained a Supply Chain Management System product that monitors real-time inventory using WebSockets.
- Implemented Firebase Realtime Database for KPI report generation around productivity metrics; integrated barcode scanner for inbound/outbound product tracking.

- Built charting views to visualise warehouse product matrices; improved application performance by optimising database schema.
- Collaborated with QA team to ensure robust testing and resolve application bugs.
- Demo: youtube.com/watch?v=Yqkp3vj2sPc

Renacle Cycling *Sep 2017 – Jun 2018*

Mobile Developer

Technologies: Android, BLE SDK, Geofencing, Push Notifications

- Contributed to implementing a BLE smart lock product used to lock/unlock cycles and integrated the BLE SDK for device-application connectivity.
- Implemented geofencing, clustering, and push notification features within the application.
- Handled customer support issues professionally and resolved application-level bugs to improve user experience.
- App link: play.google.com/store/search?q=Cymour&c=apps

EDUCATION & CERTIFICATIONS

Masters of Computer Applications (MCA)
Modern College, Pune University • First Class

Bachelor of Computer Science (BCS)
H.V. Desai College, Pune University • Higher Secondary Class

Professional Certificate in Generative AI & Machine Learning
E&ICT Academy, IIT Kanpur | 2025
LLMs, Prompt Engineering, RAG, Fine-Tuning, production GenAI system design.

Android Developer Certification
Bitcode Technologies | 2017

EXTRACURRICULAR — TECHNICAL WRITING

Published technical articles on Medium:

- medium.com/@foramadeshara
- medium.com/@foram.soni